Due: 30.11.2025

CENG311 – Programming Assignment 2
Library Management System (MIPS Assembly)

Environment: QtSpim

Objective

Design and implement a Library Management System in MIPS assembly. The project focuses on struct-like data organization, arrays, and linked lists to simulate library operations such as maintaining books, waitlists, and categories. Follow modular programming (each operation in a separate function via jal), proper stack usage, and MIPS calling convention.

You should fill in the functions in given template.

Concept Overview

Each book is represented by the following layout (offsets in bytes):

Offset	Field	Description
+0	id	Integer ID of the book
+4	title	Address of the book title string
+8	year	Publication year
+12	status	0 if available; otherwise pointer to borrower name
+16	cat_hdr	Pointer to category array header
+20	waitlist_head	Pointer to the head node of the waitlist (0 if none)

Book/Category array header layout:

Offset	Field	Description
+0	data_ptr	Pointer to allocated element area
+4	capacity	Total capacity (number of slots)
+8	count	Current number of elements

Waitlist node layout (linked list):

Offset	Field	Description
+0	name_ptr	Pointer to a string (reader name)
+4	next	Pointer to next node (0 if end)

Functional Requirements

initBookArray: Allocates a book catalog header and backing storage.

initCategoryArray: Allocates a category header and backing array.

putOnCategory: Adds a new category string to the array (unless full).

createBook: Dynamically allocates and initializes a book struct.

addBook: Adds a book pointer to the catalog array.

putOnWaitlistAt: Adds a person's name to a book's waitlist (linked list).

checkoutBookAt: Checks out the first person in the waitlist if available.

returnBookAt: Returns a book; prints proper messages depending on state.

printBookList: Iterates all books and calls printBook.

printBook: Prints title, id, year, status, categories, and waitlist.

printCategories: Prints all categories in a category header.

printWaitlist: Prints the linked list of waiting users.

Functional Requirements

Function	Arguments (Registers)	Return ($v0)	Description
initBookArray	(none)	Address of new book array header	Allocates memory for the book catalog header and its backing array (capacity = 4). Initializes data_ptr, capacity, and count. Prints 'Book list created (capacity=4, count=0)'.
initCategoryArray	(none)	Address of new category header	Allocates and initializes a category header and backing array (capacity = 4). Sets count = 0.
putOnCategory	$a0 = category header, $a1 = category string	(none)	Adds the given category string to the header's array. If full, prints 'Array full!'.
createBook	$a0 = id, $a1 = title, $a2 = year, $a3 = category header	Address of new Book struct	Allocates and initializes a new book. Sets: id, title, year, status = 0, cat_hdr = $a3, and waitlist_head = 0.
addBook	$a0 = catalog header, $a1 = book pointer	(none)	Adds a book pointer to the catalog's backing array, increments count, and prints 'Book added to list: <Title>'. Prints 'Array full!' if no space.
putOnWaitlistAt	$a0 = catalog header, $a1 = book index, $a2 = name string	(none)	Creates a new node {name_ptr, next} and appends it to the selected book's waitlist. If empty, node becomes head. Prints 'Added to waitlist: <Title> <- <Name>'.
checkoutBookAt	$a0 = catalog header, $a1 = book index	(none)	If available and waitlist not empty: removes first node, sets status = borrower name, prints 'Checked out: <Title> -> <Name>'. If no holds: 'No

			holds for: <Title>'. If already on loan: 'Already on loan: <Title> - held by <Name>'.
returnBookAt	$a0 = catalog header, $a1 = book index	(none)	Marks the book as available and prints 'Returned: <Title> <- <Name>'. If already available: 'Already available: <Title>'.
printBookList	$a0 = catalog header	(none)	Prints all books in the catalog by calling printBook for each. Precedes with '===== BOOK CATALOGUE ======'.
printBook	$a0 = book pointer	(none)	Prints the book's details (Title, ID, Year, Status) and calls printCategories and printWaitlist. Ends each entry with a separator line.
printCategories	$a0 = category header	(none)	Prints category list in the form 'Categories: c1, c2, c3'. If empty: 'Categories: (none)'.
printWaitlist	$a0 = waitlist head (node* or 0)	(none)	Prints all names in waitlist separated by ' -> '. If empty, prints 'Waitlist: (empty)'.

Programming Rules & Guidelines

Follow standard MIPS calling conventions:

• Save and restore $ra on the stack in any function that calls another function.

• Preserve $s0–$s7 registers across function calls (save on entry, restore before return).

• Temporary registers ($t0–$t9) do not need to be preserved.

Do NOT add or modify any variables in the .data section of the provided template. All the variables you need are already given in the data section.

Do NOT modify main section. You should only fill empty functions.

All data structures (book headers, arrays, waitlists) must be dynamically allocated using system calls (service 9).

Each function must be modular and reusable. Avoid merging multiple behaviors into one routine.

Ensure proper $ra saving/restoring before any nested jal calls.

Every output string must exactly match the expected format shown in the assignment output section.

Expected Output (Full Log)

```
Book list created (capacity=4, count=0)
Book added to list: Clean Code
Book added to list: Operating Systems
Book added to list: Computer Networks
Added to waitlist: Clean Code <- Alice
Added to waitlist: Clean Code <- Bob
Added to waitlist: Clean Code <- Charlie
Added to waitlist: Clean Code <- David
Added to waitlist: Clean Code <- Emma
Added to waitlist: Operating Systems <- Jane
Added to waitlist: Operating Systems <- Grace
Added to waitlist: Operating Systems <- Henry
Added to waitlist: Computer Networks <- Jack
Added to waitlist: Computer Networks <- Kate
Added to waitlist: Computer Networks <- Liam
Added to waitlist: Computer Networks <- Isla
Added to waitlist: Computer Networks <- Frank

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123  Year: 1999
Status: Available
Categories: programming, best-practices, education
Waitlist: Alice -> Bob -> Charlie -> David -> Emma
---------------------------
Title: Operating Systems
ID: 55  Year: 2014
Status: Available
Categories: systems, education
Waitlist: Jane -> Grace -> Henry
---------------------------
Title: Computer Networks
ID: 310  Year: 2011
Status: Available
Categories: networks, education
```

Waitlist: Jack -> Kate -> Liam -> Isla -> Frank

Checked out: Clean Code -> Alice
Checked out: Operating Systems -> Jane
Already available: Computer Networks

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: On Loan to Alice
Categories: programming, best-practices, education
Waitlist: Bob -> Charlie -> David -> Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: On Loan to Jane
Categories: systems, education
Waitlist: Grace -> Henry

Title: Computer Networks
ID: 310 Year: 2011
Status: Available
Categories: networks, education
Waitlist: Jack -> Kate -> Liam -> Isla -> Frank

Checked out: Computer Networks -> Jack
Already on loan: Clean Code - held by Alice
Returned: Operating Systems <- Jane

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: On Loan to Alice
Categories: programming, best-practices, education
Waitlist: Bob -> Charlie -> David -> Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: Available
Categories: systems, education
Waitlist: Grace -> Henry

Title: Computer Networks
ID: 310 Year: 2011
Status: On Loan to Jack
Categories: networks, education
Waitlist: Kate -> Liam -> Isla -> Frank

Returned: Clean Code <- Alice
Checked out: Operating Systems -> Grace
Already on loan: Computer Networks - held by Jack

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: Available
Categories: programming, best-practices, education
Waitlist: Bob -> Charlie -> David -> Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: On Loan to Grace
Categories: systems, education
Waitlist: Henry

Title: Computer Networks
ID: 310 Year: 2011
Status: On Loan to Jack
Categories: networks, education
Waitlist: Kate -> Liam -> Isla -> Frank

Returned: Computer Networks <- Jack
Checked out: Clean Code -> Bob
Already on loan: Operating Systems - held by Grace
Returned: Clean Code <- Bob

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: Available
Categories: programming, best-practices, education
Waitlist: Charlie -> David -> Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: On Loan to Grace
Categories: systems, education
Waitlist: Henry

Title: Computer Networks
ID: 310 Year: 2011
Status: Available
Categories: networks, education
Waitlist: Kate -> Liam -> Isla -> Frank

Checked out: Clean Code -> Charlie
Returned: Operating Systems <- Grace
Checked out: Computer Networks -> Kate
Returned: Computer Networks <- Kate
Checked out: Operating Systems -> Henry

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: On Loan to Charlie
Categories: programming, best-practices, education
Waitlist: David -> Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: On Loan to Henry
Categories: systems, education
Waitlist: (empty)

Title: Computer Networks
ID: 310 Year: 2011
Status: Available
Categories: networks, education
Waitlist: Liam -> Isla -> Frank

Returned: Clean Code <- Charlie
Checked out: Clean Code -> David
Checked out: Computer Networks -> Liam

Returned: Operating Systems <- Henry
No holds for: Operating Systems

===== BOOK CATALOGUE =====
Title: Clean Code
ID: 123 Year: 1999
Status: On Loan to David
Categories: programming, best-practices, education
Waitlist: Emma

Title: Operating Systems
ID: 55 Year: 2014
Status: Available
Categories: systems, education
Waitlist: (empty)

Title: Computer Networks
ID: 310 Year: 2011
Status: On Loan to Liam
Categories: networks, education
Waitlist: Isla -> Frank

Deliverables

• PA2_StudentID.asm (your assembly source file).

Submission & Deadline

Submit via Microsoft Teams.